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Palaeomagnetism of Ordovician and Silurian rocks from the Chu-Yili and Kendyktas mountains, south Kazakhstan

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SUMMARY

Despite significant progress during the last decade, the overwhelming majority of the details of the crustal evolution of Kazakhstan remain enigmatic. The central question of whether Kazakhstan amalgamated during the Ordovician and has acted as a coherent microplate since or whether Kazakhstan is the result of continuous accretion of terranes and arcs along a major subduction system throughout the Palaeozoic is still unanswered. Defining structural units and reconstructing their interaction relative to each other and with respect to Siberia and Baltica will improve our understanding of continent formation in Central Asia. Addressing this problem, a detailed palaeomagnetic study of the Palaeozoic formations in the Chu-Yili and Kendyktas mountains, south Kazakhstan was undertaken.

Detailed stepwise thermal demagnetization experiments reveal the presence of a post-folding component of magnetization with intermediate to high unblocking temperatures (component A) in almost all samples. The *in situ* mean directions identified in 31 sites range in declination between 150° and 260° and show intermediate and upward pointing inclinations; based on the mean inclination a late Palaeozoic to early Mesozoic age is assumed for this secondary component of magnetization.

After removal of component A a second component of magnetization (component B) was identified in red coloured argillites of early Arenigian age (Kendyktas Mountains) as well as in sediments of Early Silurian and Late Silurian to Early Devonian age (Chu-Yili Mountains). Component B passes the fold test (Enkin 2003) on the 95 per cent confidence level and yields a bedding-corrected site mean direction of $D = 9.2°$, $I = -16.9°$ (five sites, $k = 26.9$, $\alpha_{95} = 15.0°$) for the Lower Ordovician unit and $D = 346.9°$, $I = +23.8°$ (eight sites, $k = 20.8$, $\alpha_{95} = 12.4°$) for the Silurian and Lower Devonian units. Based on the age of angular unconformities, the timing of regional deformation in the Agalatas area can be constrained to be Late Ordovician in age and Middle Devonian in the Chu-Yili area. Based on the mean inclination, a southerly palaeolatitude of 8.6°–8.7°/+ 7.6° can be derived for the study area during the Early to Late Ordovician and a northerly palaeolatitude of 12.4° + 7.7°/− 6.6° for the Silurian to Early Devonian. The observed mean declinations for the post-folding component A are significantly different from the expected reference directions for the Permian based on either the apparent polar wander path for Baltica or Siberia and suggest counter-clockwise rotations of Kazakhstan with respect to both during Devonian to Permian time.

Key words: Lower Ordovician, palaeomagnetism, Silurian, south Kazakhstan.

1 INTRODUCTION

The northern part of Eurasia as of today includes the Baltica, Siberia, Kazakhstan, Turan and Tarim palaeocontinents which amalgamated during the Carboniferous and Permian. Within this agglomerate of continents and microcontinents, Kazakhstan occupies a central position (Fig. 1a) and represents a key area for understanding the processes of amalgamation of Eurasia during the Palaeozoic.

Unlike the Precambrian cratons of Baltica, Siberia and Tarim, Kazakhstan can be described as an agglomerate of Proterozoic and Lower Palaeozoic microcontinents, Lower and Middle Palaeozoic island arcs and ophiolite belts (Avdeyev 1990; Yakubchuk 1990;

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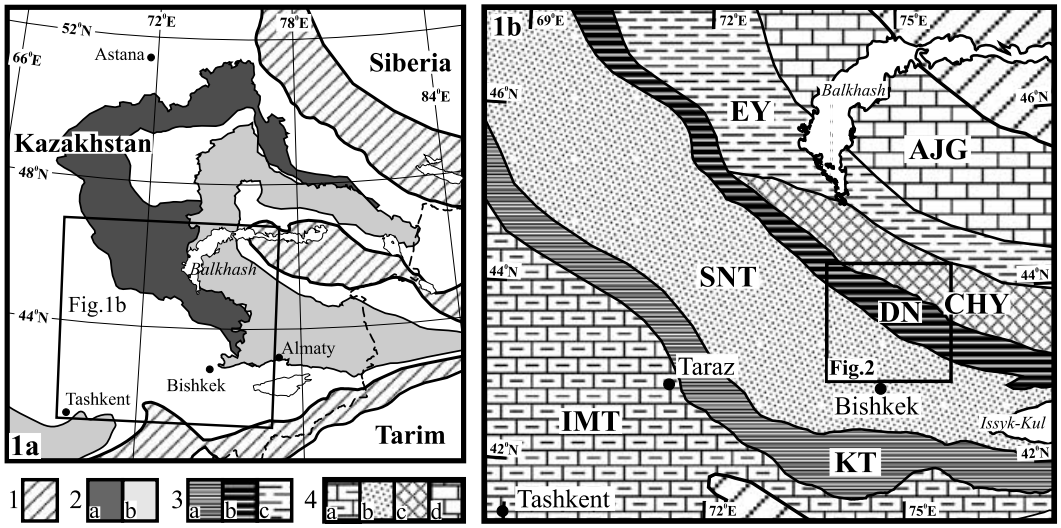


Figure 1. Tectonic setting of Upper Palaeozoic Kazakhstan (1a) and Lower Palaeozoic south Kazakhstan and north Tian-Shan (1b). 1, Upper Palaeozoic thrust folded belts; 2, subductional(?) volcanic belts: (a) Lower and Middle Devonian and (b) Carboniferous and Permian; 3, Lower Palaeozoic suture zones: (a) Kyrgyz–Terskey (KT), (b) Dzhalairy–Naiman (DN) and (c) Erementau–Yili (EY); 4, microcontinents: (a) Ishim–mid-Tian-Shan (IMT), (b) Stepnyak–north Tian-Shan (SNT), (c) Chu-Yili (CHY), (d) Atasu–Junggar (AJG).

Sengör *et al.* 1993). However, deciphering the details of the processes which finally led to the formation of Kazakhstan is rather difficult due to the very complicated regional structural patterns and scarcity of reliable palaeomagnetic data. The controversial points in which these models (Zaytsev 1984; Avdeyev 1990; Yakubchuk 1990; Zonenshain *et al.* 1990; Mossakovsky *et al.* 1993; Sengör *et al.* 1993; Apollonov 2000) differ are centred around the questions of how to (a) identify the individual terranes, (b) quantify their drift histories and (c) explain the geodynamic processes, ultimately leading to the bending of Palaeozoic fold belts as observed today. Addressing these problems, a detailed palaeomagnetic study has been carried out in the Chu-Yili and Kendyktas mountains (Figs 1 and 2). This area is located in the central part of Kazakhstan where red beds as well as volcanic suites are well exposed. The rocks range in age from the Early Ordovician to the Carboniferous and are exposed in well-defined fold structures. Here we present data on the palaeomagnetism of the Ordovician and Silurian rocks from the Chu-Yili and Kendyktas mountains.

2 TECTONIC SETTING

The lower Palaeozoic basement of southern Kazakhstan and the northern Tian-Shan comprises four microcontinents, which are separated one from another by tectonic suture zones (Fig. 1b). Four microcontinents, Ishim–Middle Tian-Shan (IMT), Stepnyak–north Tian-Shan (SNT), Chu-Yili (CHY), and Atasu-Junggar (AJG) can be identified as elongated blocks within the Lower Proterozoic crystalline basement and the Upper Proterozoic and/or Lower Palaeozoic sedimentary cover. The intervening suture zones, Kyrgyz–Terskey (KT), Dzhalair–Nayman (DN) and Erementau–Yili (EY), are marked by Lower Palaeozoic ophiolites and deep marine sediments. Within the KT suture the rocks range in age from the Early Cambrian to Middle Arenigian; in the DN suture from Early Cambrian(?) to Tremadocian, and in the YE suture zone they vary in age from the Middle Cambrian to Caradocian (Abdulin *et al.* 1980; Avdeyev & Kovalev 1989; Mikolaichuk *et al.* 1997; Apollonov 2000; Nikitin & Nikitina 2000).

The DN and KT suture zones mark the amalgamation of the SNT, CHY and apparently the IMT microcontinents during the beginning and the middle of the Arenigian age, respectively (Avdeyev 1990; Mikolaichuk *et al.* 1997). During Late Arenigian and Llanvirnian times this composite microcontinent was dominated by passive margin deposits. From the Llandeilian onwards up to the Ashgillian the tectonic setting of this terrane was changing and the Stepnyak–north Tian-Shan volcanic arc evolved above a southwest-dipping (today's coordinates) subduction zone. The volcanic arc is located mainly within the former SNT and KT tectonic domains, whereas the forearc basin includes the former CHY and DN terranes (Apollonov & Patalakha 1989; Mikolaichuk *et al.* 1997; Apollonov 2000). The EY deeper marine basin developed from Cambrian to the Late Ordovician and was eliminated due to the collision of the Atasu-Junggar microcontinent with the Stepnyak–north Tian-Shan arc (Apollonov 2000). As a result of this collision the activity of the volcanic arc ceased, almost the entire area from AJG to IMT experienced substantial deformation and granite batholiths were emplaced along the whole length of the former arc. However, significant deformation did not occur in the Chu-Yili area, where sedimentation continued during the Early Silurian.

3 GEOLOGY AND SAMPLING

A palaeomagnetic study has been conducted (a) in the Kendyktas Ridge and (b) in the south of the Chu-Yili Mountains (Fig. 2). The first area is a part of the Stepnyak–north Tian-Shan facies zone of Ordovician age (Nikitin 1972) and the second belongs to the Selety–Chu-Yili facies zone of Ordovician and Silurian ages (Bandaletov 1969; Nikitin 1972). A detailed description of the Palaeozoic stratigraphy of these areas is given in a number of publications (Keller & Rukavishnikova 1961; Nikitin 1972; Abdulin *et al.* 1980; Nikitin *et al.* 1991a,b; Popov *et al.* 2001) and is only summarized here.

In the Kendyktas Ridge the Lower Palaeozoic rocks range in age from Late Cambrian to Ashgillian. Upper Cambrian and Lower Ordovician rocks are exposed in the south of the ridge with the best sections located on the banks of Agalatas river (Fig. 2a, area 1). The section is subdivided into three formations, which are dominated by

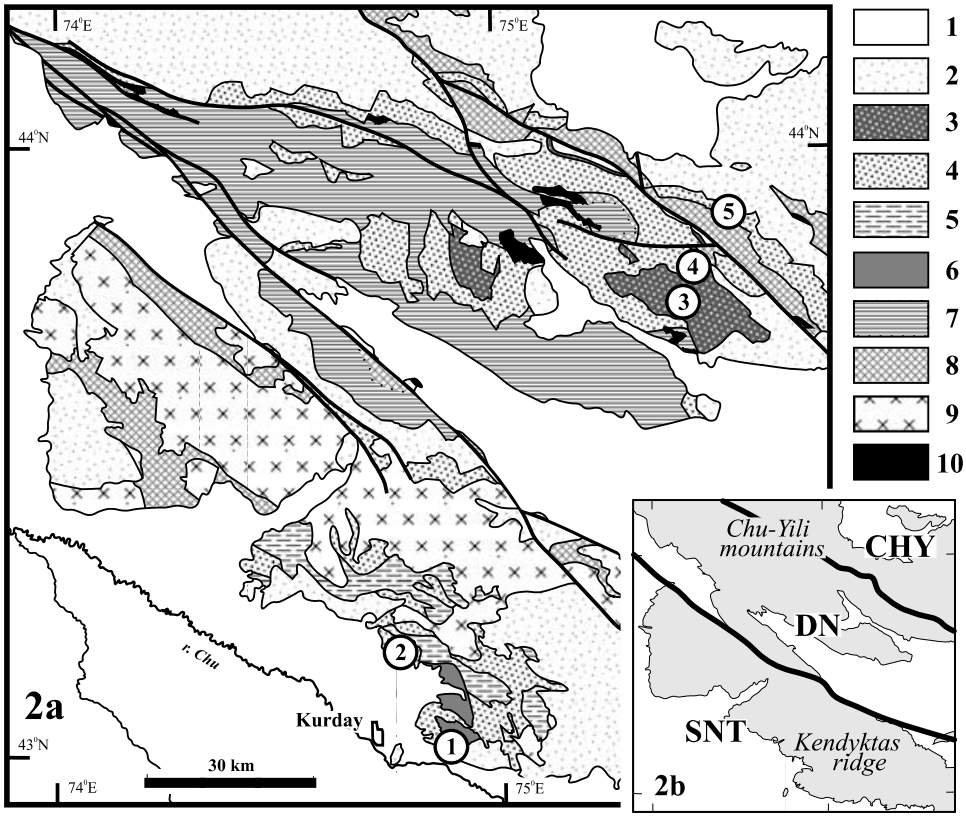


Figure 2. Schematic geological map of the south Chu-Yili and Kendyktas mountains (modified after Myasnikov *et al.* 1979) (2a). Orographic and structural patterns (2b). 1, Cenozoic; 2, Devonian and Upper Palaeozoic; 3, Silurian; 4, Middle and Upper Ordovician; 5, Lower and Middle Ordovician; 6, Lower Ordovician; 7, Cambrian and Lower Ordovician; 8, Proterozoic; 9, Upper Ordovician granites; 10, ophiolites. Numbers in circles show sampling areas: 1, Agalatas river (AGA); 2, Georgievka (GEO); 3, 4, Dulankara Mountains: 3, sites DUL 1–8; 4, sites DUL 9–11, 15–19; 5, Anderkenyn–Akchoku ravine (AND). Abbreviations: SNT, Stepnyak–north Tian-Shan microcontinent; CHY, Chu-Yili microcontinent; DN, Dzhalair–Naiman suture zone.

shallow marine deposits. They consist from bottom to top of brown, green and grey sandstones and siltstones (Kendyktas formation, Upper Cambrian and Lower Tremadocian), light-coloured massive thick-bedded fossiliferous limestones (Agalatas formation, Upper Tremadocian) and red-pink and violet sandstones, siltstones and limestones (Kurday formation, Lower Arenigian). Ages of all three formations are determined with a precision of a substage based on numerous brachiopod and trilobite fossils (Keller & Rukavishnikova 1961; Rukavishnikova & Salin 1965; Nikitin 1972; Nikitin *et al.* 1991a,b; Popov *et al.* 2001).

Rocks of the Upper Arenigian and Middle and Upper Ordovician are well exposed on the southwestern slopes of Kendyktas (Fig. 2). Here, the Shcherbakty formation (Upper Arenigian and Llanvirnian) consists of green and grey-coloured argillites, sandstones and shales, which accumulated in passive margin depositional setting. The overlying formations were formed within a volcanic island arc (Nikitin *et al.* 1991a,b; Ergaliev *et al.* 1995; Apollonov 2000). The Rgaity formation (Llandeilian and Lower Caradocian) consists of shales, tuffs and intermediate volcanic rocks in the lower part of the section, and is dominated by red-coloured massive and cross-bedded sandstones and siltstones (about 850 m thick) in the upper part. The Keskintas formation (Upper Caradocian) comprises andesitic and basaltic porphyrites, tuffs, green sandstones, limestones and conglomerates. Up-section it is changing into the brown and green conglomerates, sandstones and siltstones of the Taspaly formation, which is dated tentatively as the Upper Ordovician (Nikitin 1972; Nikitin *et al.* 1991a,b).

Two Early Palaeozoic deformational events of mid-Arenigian and Late Ordovician age can be identified in the Kendyktas ridge. The mid-Arenigian event, which is coeval with an episode of suturing within the KT zone (Mikolaichuk *et al.* 1997) resulted only in a low-angle unconformity without major deformation. Main deformations occurred in the Late Ordovician, when Cambrian and Ordovician rocks were tightly folded with northwest–southeast striking fold axes. The age of this event is based on the fact that the youngest rocks affected by folding are Late Ordovician in age, and that the granite batholiths cutting the folds are Late Ordovician and Silurian in age (466 to 438 Myr according to $^{207}Pb/^{206}Pb$ isotope data) (Myasnikov *et al.* 1979; Mikolaichuk *et al.* 1997). This main deformation event can be correlated with the collision of the AJG microcontinent against the SNT volcanic arc (Apollonov 2000).

In the Chu-Yili Mountains the Lower Palaeozoic formations can be clearly divided into two complexes. The lower one developed within the DN suture zone and comprises Cambrian to Tremadocian ophiolites and deep marine sediments. These rocks are typically intensely deformed and metamorphosed. The upper complex consists predominantly of siliciclastic rocks, ranging in age from the Arenigian to the Silurian. They unconformably overlay both the Cambrian and Lower Ordovician rocks of the DN suture and the Lower Proterozoic crystalline schists within the CHY microcontinent.

The Arenigian and Llanvirnian comprise shallow marine siliciclastics, which contain a minor amount of andesites in the Arenigian stage. Llandeilian, Caradocian and Lower Ashgillian rocks include a wide variety of siliciclastic deposits, ranging from

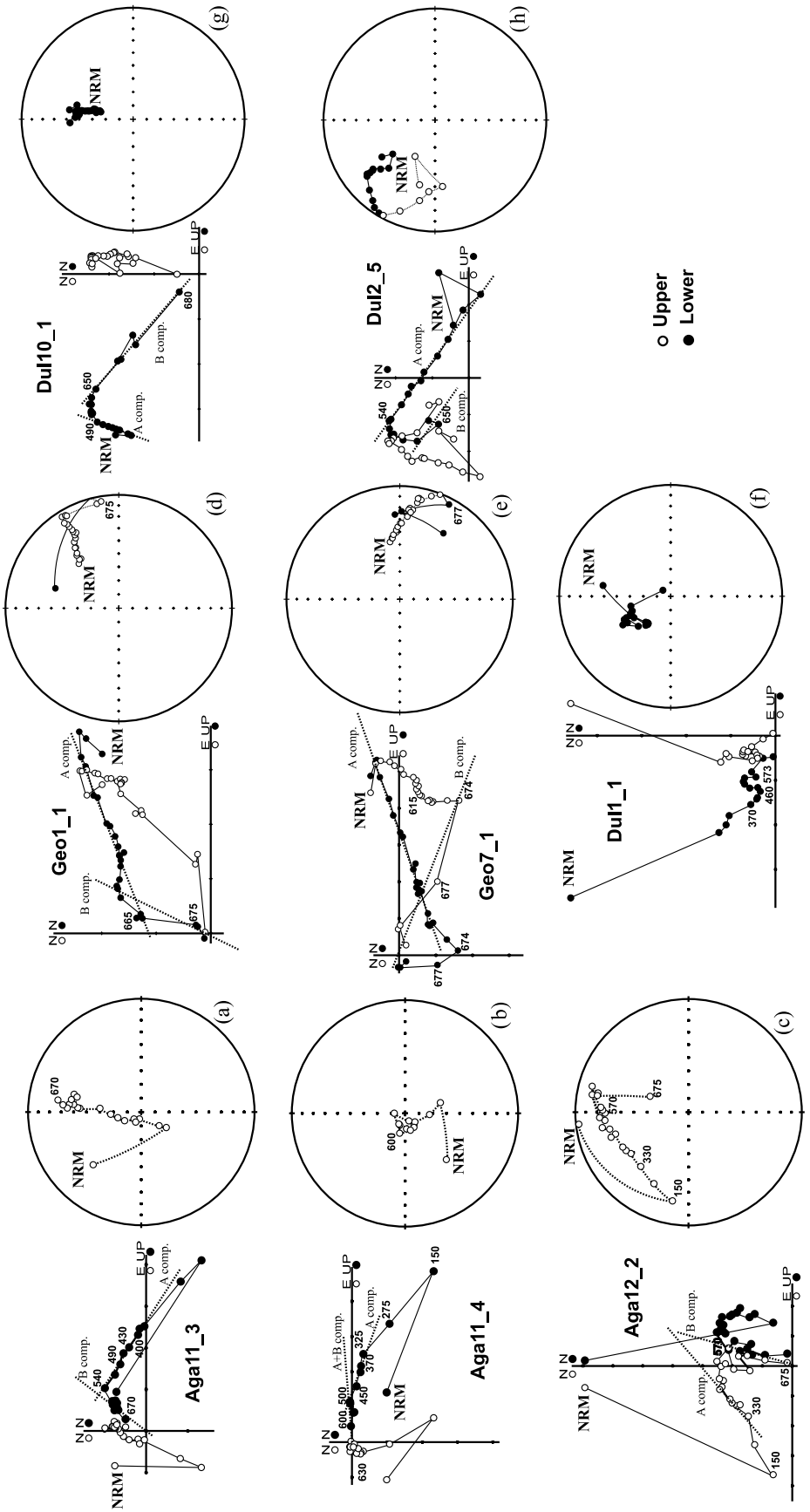


Figure 3. Orthogonal plots of thermal demagnetization data. Temperatures are in °C. Stratigraphic coordinates.

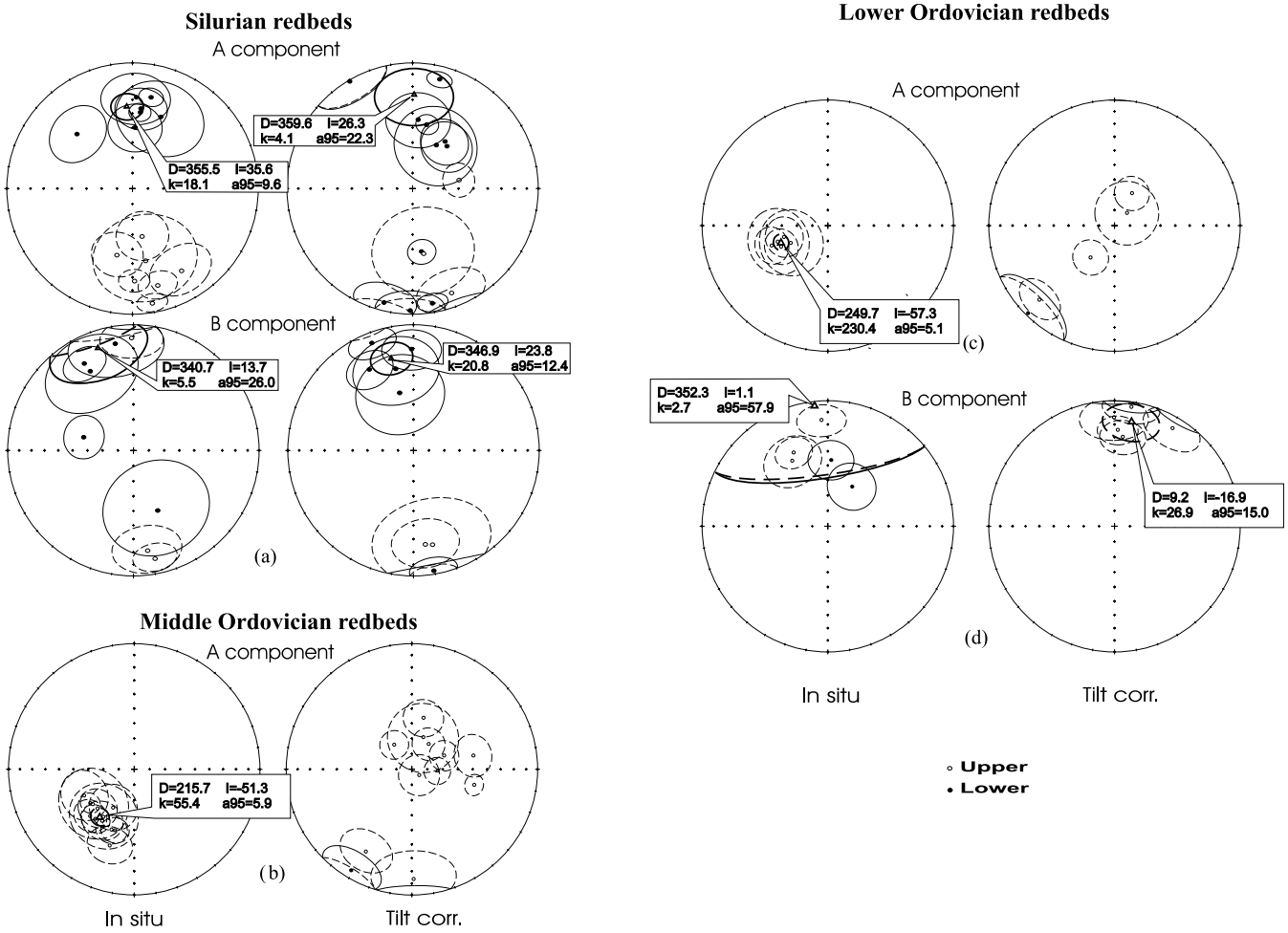


Figure 4. Site-mean directions and α_{95} confidence circles. Open (solid) symbols = upper (lower) hemisphere. The mean directions are listed in Table 1.

non-marine and shallow marine facies to deeper marine turbidites (Abdulin *et al.* 1980; Popov *et al.* 2001). These deposits are interpreted to have been formed in the forearc basin of the SNT arc (Apollonov 2000).

The uppermost Ordovician and Silurian formations were accumulated in a nearshore part of the Junggar–Balkhash Basin, which was situated northeast of the Chu-Yili area. The Ashgillian and Lower Llandowerian (Dzhalair and Salamat formations) consist of green- and grey-coloured marine sandstones, siltstones and argillites (Apollonov *et al.* 1988). They are disconformably overlain by light-coloured massive and cross-bedded quartz-rich sandstones and conglomerates of the Lower Silurian Betkainar formation. The last one changes up-section into the red-coloured sandstones and siltstones of the Koiche formation. The Early Silurian age of the Koiche formation is based on brachiopods collected from limestone strata in the lower part of the formation (Koren *et al.* 2003).

The Koiche formation is overlain by non-marine red-coloured sandstones and conglomerates, which change further up-section into the Lower Devonian rocks. The red sandstone unit can be dated as Late Silurian to Early Devonian according to its stratigraphic position between Lower Silurian and Lower Devonian (Abdulin *et al.* 1980). Occasionally, a minor low-angle unconformity (5–15°) between Lower Silurian and Lower Devonian rocks can be identified in several localities in the Chu-Yili Mountains. This, however, is the exception and generally the Silurian and Lower and Middle Devonian rocks seem to have been deformed conformably and form coherent fold structures. These folds are cut by granite intrusions of Late Givetian and Frasnian age bracketing the time of folding to be Givetian (Abdulin *et al.* 1980; Zaytsev 1984).

In the Kendyktas region, palaeomagnetic samples were collected from the lower parts of the Kurday formation of the early Arenigian age along the Agalatas river some 15 km east of Kurday (Fig. 2, area 1, locality AGA), and from the Llandeilian and Caradocian red beds of the Rgaity formation 20 km to the north of Kurday (Fig. 2, area 2, locality GEO). In the south Chu-Yili region palaeomagnetic sampling concentrated on the Dulankara mountains and the Anderkenyn-Akchoku ravine. The samples were taken from red sandstones of the Lower Silurian Koiche formation (Fig. 2, area 3, sites DUL 2–8; area 4, sites DUL 15–19), and from the Upper Silurian and Lower Devonian sandstones (Fig. 2, area 4, sites DUL 9–11; area 5, locality AND).

4 METHODS AND RESULTS

Each unit has studied has been sampled at 5 to 13 sites (five to seven drill cores at each site). All samples (1 inch in diameter) have been taken using a gasoline-powered drill and oriented using a standard magnetic compass. One (sometimes two) specimen(s) from each core was subjected to stepwise thermal demagnetization in 15 to 30 steps up to 690 °C in a Schonstedt oven and measured

Table 1. Site mean directions: *D*, declination; *I*, inclination; *k*, precision parameter (Fisher 1953); α_{95}, radius of confidence circle.

Site	Number of specimens	*In situ*				Tilt corrected			
		D	*I*	*k*	α_{95}	*D*	*I*	*k*	α_{95}
Lower Ordovician (locality AGA) (component A)									
AGA 9	6	245.5	−56.4	36.7	11.2	28.4	−66.2	36.7	11.2
AGA 10	6	252.0	−55.5	11.5	20.7	44.7	−78.6	11.5	20.7
AGA 11	6	250.3	−50.7	34.2	11.6	216.3	−63.9	35.1	11.5
AGA 12	6	255.3	−60.3	28.8	12.7	225.4	−16.9	28.7	12.7
AGA 13	5	244.7	−63.3	14.1	21.1	224.8	2.4	14.1	21.1
Mean		**249.7**	**− 57.3**	**230.4**	**5.1**	**226.8**	**− 59.3**	**2.7**	**57.3**
Lower Ordovician (locality AGA) (component B)									
AGA 9	6	331.4	−40.5	12.6	17.7	359.7	−14.5	12.6	17.7
AGA 10	4	335.1	−35.6	69.6	11.1	1.9	−24.3	69.6	11.1
AGA 11	4	356.5	−16.6	123.1	8.3	5.2	−29.2	51.9	12.9
AGA 12	6	2.7	46.0	24.7	13.8	8.0	−4.6	24.7	13.8
AGA 13	5	31.1	59.6	25.4	15.5	30.6	−10.6	25.4	15.5
Mean		**352.3**	**1.1**	**2.7**	**57.9**	**9.2**	**− 16.9**	**26.9**	**15.0**
Middle Ordovician (locality GEO) (component A)									
GEO 1	6	223.2	−47.5	40.9	10.6	33.6	−70.4	40.9	10.6
GEO 2	6	197.5	−47.6	60.3	8.7	93.9	−73.5	60.3	8.7
GEO 3	4	218.0	−48.3	98.7	9.2	324.1	−70.4	98.7	9.2
GEO 5	5	197.8	−36.3	34.0	13.3	129.0	−84.0	34.0	13.3
GEO 6	4	223.0	−54.7	53.9	12.6	12.7	−55.5	53.9	12.6
GEO 7	7	200.6	−47.3	39.9	9.7	67.1	−67.3	39.9	9.7
GEO 9	3	208.7	−61.3	65.9	15.3	209.2	−25.8	65.9	15.3
GEO 10	4	212.9	−52.6	25.8	18.5	179.1	−13.6	18.7	21.8
GEO 14	4	242.1	−52.8	34.9	15.8	211.2	7.0	34.9	15.8
GEO 16	6	211.8	−49.9	27.7	13.0	77.5	−48.5	27.7	13.0
GEO 16–1	6	229.7	−54.4	9.6	22.8	19.6	−68.0	9.6	22.8
GEO 17	6	232.3	−53.5	106.9	6.5	104.2	−47.6	106.9	6.5
Mean		**215.7**	**− 51.3**	**55.4**	**5.9**	**130.0**	**− 78.6**	**3.4**	**28.1**
Silurian and Lower Devonian (localities DUL and AND) (component A)									
DUL 2	7	2.4	49.7	14.4	16.5	39.3	54.5	14.4	16.5
DUL 3	4	6.2	36.1	37.5	15.2	34.5	52.7	37.5	15.2
DUL 5	3	2.0	28.5	41.5	19.4	4.7	44.0	41.5	19.4
DUL 6	7	11.5	26.7	77.9	6.9	12.9	46.6	77.9	6.9
DUL 7	5	21.6	38.6	9.3	26.4	28.5	57.6	9.3	26.4
DUL 8	6	169.7	−41.2	5.6	31.2	170.5	−46.1	5.6	31.2
DUL 9	6	192.9	−44.7	19.3	15.6	193.8	7.3	19.3	15.6
DUL 10	6	168.2	−58.0	17.4	16.5	181.1	3.1	17.4	16.5
DUL 11	4	179.0	−27.2	115.5	8.6	171.8	47.6	115.5	8.6
DUL 15	5	170.2	−8.4	99.1	7.7	170.1	8.5	99.1	7.7
DUL 16	5	149.4	−24.8	18.9	18.1	159.5	−12.3	18.9	18.1
DUL 17	3	5.6	38.9	359.6	6.5	13.9	11.6	359.6	6.5
DUL 18	6	165.3	−21.4	40.0	10.7	79.7	−59.3	40.0	10.7
AND 4	5	314.9	37.8	18.5	18.3	329.6	0.5	18.5	18.3
Mean		**355.5**	**35.6**	**18.1**	**9.6**	**359.6**	**26.3**	**4.1**	**22.3**
Silurian and Lower Devonian (localities DUL and AND) (component B)									
DUL 2	7	331.3	29.0	5.8	27.5	346.5	51.2	5.8	27.5
DUL 4	5	359.0	−11.6	24.2	15.9	358.9	20.4	24.2	15.9
DUL 7	4	350.4	15.0	14.5	25.0	347.5	34.0	14.5	25.0
DUL 8	6	172.2	−20.4	13.7	18.7	172.7	−25.3	13.7	18.7
DUL 15	5	168.6	−13.7	45.9	11.4	170.5	3.0	45.9	11.4
DUL 16	6	330.5	21.7	30.8	12.3	338.9	9.1	30.8	12.3
DUL 18	6	158.4	47.2	5.1	33.0	168.6	−24.2	5.1	33.0
AND 4	6	285.0	56.3	25.2	13.6	331.4	27.5	25.2	13.6
Mean		**340.7**	**13.7**	**5.5**	**26.0**	**346.9**	**23.8**	**20.8**	**12.4**

with a 2G cryogenic magnetometer. All equipment used is housed in a magnetically shielded room in the Geophysics Section of the Department for Earth and Environmental Sciences, Munich. Demagnetization results were analysed on orthogonal vector diagrams (Zijderveld 1967) using the palaeomagnetic software written by Randy Enkin. The directional characteristics of almost all samples are controlled by the presence of two distinguishable components of magnetization.

Component A is based on at least four consecutive demagnetization steps (Fig. 3a). Component A is normally removed during stepwise heating at temperatures of up to 540 °C and is characterized by intermediate to steep upward-pointing inclinations and

Table 2. Obtained magnetic directions and virtual geomagnetic pole positions.

Locality	Age of rocks	Loc. lat.	Loc. long.	*D*	*I*	*k*	α_{95}	Pol. lat	Pol. long.	*dp*/*dn*	φ	Age of magnetization
DUL, AND	Silurian and Lower Devonian (Post-folding comp.)	43°48′	75°28′	355.5	35.6	18.1	9.6	65.6	265.8	6.4/11.1	19.7 +7.3/−6	Post-Middle Devonian
DUL, AND	Silurian and Lower Devonian (Pre-folding comp.)	43°48′	75°28′	346.9	23.8	20.8	12.4	56.7	279.3	7.1/13.4	12.4 +7.7/−6.6	Silurian to Early Devonian
GEO	Middle Ordovician	43°10′	74°53′	215.7	−51.3	55.4	5.9	59.8	174.6	5.4/8.0	32.0 +5.8/−5.1	Post-Late Ordovician
AGA	Lower Ordovician (Post-folding comp.)	43°02′	74°54′	249.7	−57.3	230.4	5.1	38.3	145.3	5.4/7.4	37.9 +5.8/−5.1	Post-Late Ordovician
AGA	Lower Ordovician (Pre-folding comp.)	43°02′	74°54′	9.2	−16.9	26.9	15.0	37.7	243.3	8.0/15.5	−8.6 −8.7/+7.6	Early to Late Ordovician

Loc. lat. and Loc. long, the locality position; D, declination; I, inclination; k, precision parameter (Fisher 1953); α_{95}, radius of confidence circle. Pol. lat., Pol. long. and *dp*/*dn* are the latitude, longitude and radius of 95 per cent confidence circle, respectively, of the palaeopole; φ, palaeolatitude.

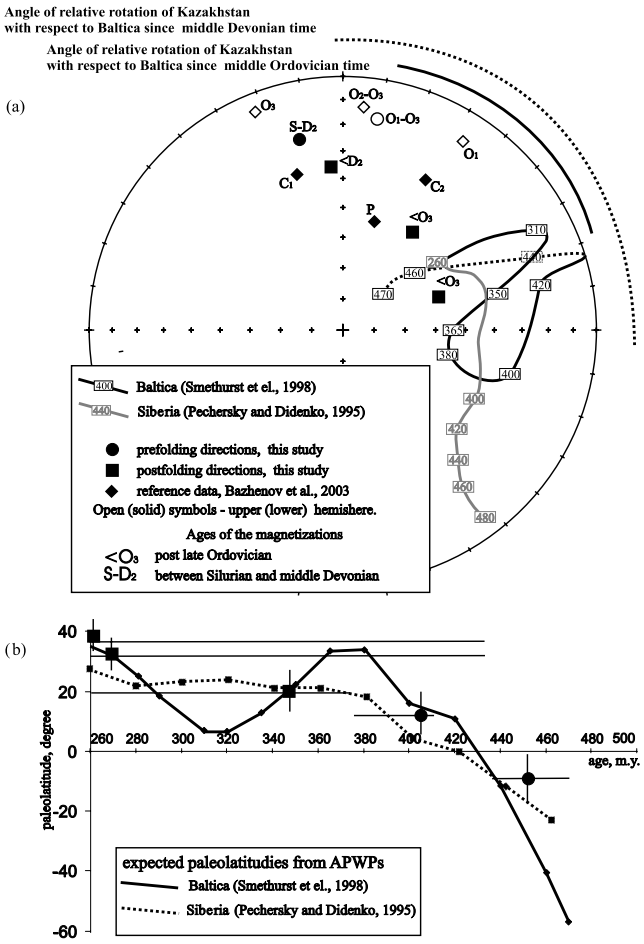


Figure 5. Observed and reference directions for the north Tian-Shan and Baltica: (a) comparison of directions, (b) comparison of palaeolatitudes.

declinations between 215° and 355° in geographical coordinates (Fig. 3). Only occasionally are the maximum unblocking temperatures in excess of 580 °C indicating that in addition to magnetite, haematite might also be a carrier of this magnetization. Although the majority of component A is of single (reversed) polarity, we note that in samples from localities DUL and AND normal and reverse directions of component A can be observed. Component A fails the fold test (Enkin 2003) and is clearly of secondary origin. Significant differences in declination and inclination as well as the occurrence of normal and reversed polarity can be used to argue against a short time interval of remagnetization. It is therefore argued that the various units have been remagnetized at different points in time ranging from the mid-Devonian to the Permian.

Component B identified in the Lower Ordovician rocks (the Kendyktas area) sometimes displays very narrow unblocking temperature spectra (about 5 °C) and is isolated in the temperature interval 580–680 °C (Fig. 3c), indicating haematite to be the carrier. In several samples the high-temperature component isolated shows close similarity to the direction of component A (Fig. 3b). The distribution of component B significantly improves upon tilt correction (Fig. 4d) and passes the fold test (Enkin 2003) on the 95 per cent confidence level. The resulting mean direction ($D = 9.2°$, $I = -16.9°$, $k = 26.9$, $\alpha_{95} = 15.0°$) is therefore considered as being pre-folding in age. The age of folding in the Agalatas area is defined as being Late Ordovician(see Geological description). Based on the negative fold test a post-deformational age, i.e. post-Late Ordovician can be derived for component A, whereas the positive outcome of the fold test for component B demonstrates that in the Kendyktas area this component is unambiguously of pre-folding in age and is very likely to have been acquired during the Early to Late Ordovician.

Component B for the Silurian–Early Devonian rocks (Chu-Yili area) was isolated in the temperature interval 580–680 °C (Figs 3g and h), again as above, indicating haematite to be the carrier. Again, the distribution of the site-mean directions of component B improves significantly upon stratigraphic correction (Fig. 4a) and passes the fold test (Enkin 2003) on the 95 per cent confidence level. The resulting direction ($D = 346.9°$, $I = 23.8°$, $k = 20.8$, $\alpha_9 5 = 12.4°$) is therefore of pre-folding origin.

Based on the positive fold test, component B is interpreted to be pre-folding in origin, and thus pre-dates the mid-Devonian deformational event.

5 PALAEOGEOGRAPHIC IMPLICATION

The resulting palaeolatitude for component B based on a mean direction with a declination of 9.2° and inclination of −16.9° puts the SNT into a southerly palaeolatitude of 8.6°–8.7°/+ 7.6° if the normal polarity option is chosen. This result is in agreement with both the data obtained for the Middle and Upper Ordovician rocks in the northern Tian-Shan (Bazhenov *et al.* 2003) (Fig. 5), and with the data reported for Ordovician ophiolites in central Kazakhstan (Grishin *et al.* 1991; Pechersky & Didenko 1995) all reporting northerly declinations for presumably Ordovician magnetizations. If

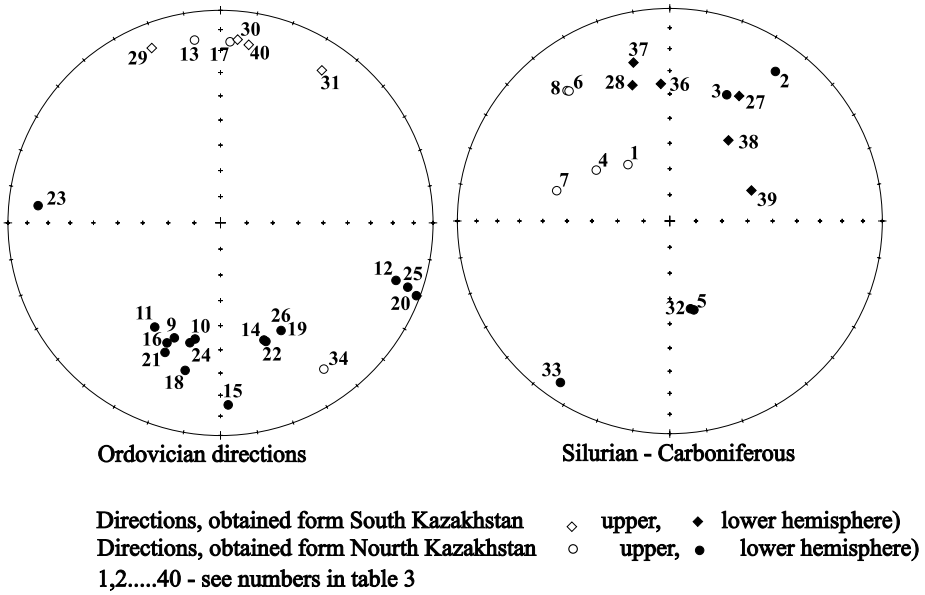


Figure 6. Comparison of obtained directions from north and central Kazakhstan to south Kazakhstan.

these results are correct, they imply no major rotations between the northern Tian-Shan, including the Agalatas area and central Kazakhstan since the Ordovician. However, the quality of the data from central Kazakhstan remains questionable (Fig. 6). The age of magnetization is not well constrained and it is rather debatable to what extent palaeomagnetic data from ophiolites can be taken as being representative for microcontinents and/or terranes.

The pre-folding component of magnetization isolated in the Silurian and Lower Devonian rocks in the south Chu-Yili Mountains shows a northerly declination and intermediate positive inclinations ($D = 346.9^\circ$, $I = 23.8^\circ$) and indicates a northerly palaeolatitude of the area of $12.4^\circ \pm 7.7^\circ$ if again a normally polarity is assumed. This result, however, is in sharp contrast to other data published earlier for Silurian and Devonian rocks in central Kazakhstan (Grishin *et al.* 1997). These show southerly declinations, which differ from the results of this study by up to 150°. However, this cannot be taken as evidence for oroclinal bending. The area to the northwest of Balkhash, where a pre-folding magnetization of Devonian age ($D = 319^\circ$, $I = -52^\circ$) has been reported (Grishin *et al.* 1997) and the Chu-Yili Mountains are located within a structurally continuous belt with uniform strike. Therefore, major rotations seem unlikely. In addition, the fact that the Ordovician directions for central Kazakhstan and the northern Tian-Shan agree is a strong argument against major oroclinal bending. Nevertheless, more high-quality data are urgently needed before this matter can be finally decided.

Ordovician magnetizations from the Chingiz Range (north Kazakhstan) differ significantly from those reported here. Whereas only northerly declinations have been identified in the Ordovician rocks from the northern Tian-Shan and central Kazakhstan, results from coeval rocks from the Chingiz Range consistently show southerly declinations (Collins *et al.* 2003) (see also Table 3). These differences in declination are compatible with a large-scale change in strike between the Chingiz range and the northern Tian-Shan and can be taken as evidence for oroclinal bending between the Chingiz and northern Tian-Shan (Collins *et al.* 2003; Levashova *et al.* 2003). Unfortunately, however, the data published by Collins *et al.* (2003) for the Chingiz Range shows some substantial scatter making it difficult to draw firm conclusions.

Minimizing drift velocities and going back in time from accepted palaeogeographic reconstructions for Permian times, shallow northerly palaeolatitudes in the Silurian and Devonian and shallow southern palaeolatitudes in the Ordovician seem rather reasonable and suggest that our assumption that the Ordovician magnetization was acquired during periods of normal polarity of the Earth' magnetic field is justified (Fig. 5a) (see Table 2 and 3).

All obtained data indicate that south Kazakhstan has good latitude agreement with Baltica from the Ordovician to the Late Palaeozoic, but was slowly rotated (Fig. 5b). A further discussion of palaeogeographic details of south Kazakhstan with respect to Baltica is beyond the scope of this paper. This is equally true for a comparison between south Kazakhstan (north Tian-Shan Block) and north Kazakhstan (Chingiz and central Kazakhstan). Both issues can only be addressed in future on the basis of much more data of reliable quality.

6 ROTATIONS RELATIVE TO BALTICA AND SIBERIA SINCE THE ORDOVICIAN TO PERMIAN

Both the post-folding component A (mid-Devonian to Permian in age) and the pre-folding component B (Ordovician and Silurian/mid-Devonian in age) show declinations which are significantly different from the relevant reference directions derived from Baltica (Smethurst *et al.* 1998) and Siberia APWP (Pechersky & Didenko 1995). These differences indicate counter-clockwise rotations of south Kazakhstan with respect to Baltica and Siberia of up to 80° and 140°, respectively (Fig. 5a). The upper age limit of these rotations is not clearly defined but a Permian age is rather likely (Alexyutin *et al.* 2003). The declinations for component A indicate counter-clockwise rotations of Kazakhstan with respect to both continents since the mid-Devonian.

7 CONCLUSION

(1) Pre-folding magnetizations were isolated: (a) for the Lower Ordovician rocks in the Kendyktas ridge ($D = 9.2^\circ$, $I = -16.9^\circ$, $k =$

Table 3. Reference palaeomagnetic direction for the rectangle 41°E to 52°E and 70°N to 80°N (only for Kazakhstan). Selected from the Global Palaeomagnetic Database (McElhinny 1998 version 4.4).

Mo. ref. no.	Authors	Database	Age	Place	PLat.	PLong.
1	Pechersky & Didenko (1995)	3045	377–391	Maikaner Mulda, central Kazakhstan	−4.9	279.2
2	Rusinov (1986)	2025	363–417	Central Kazakhstan	43.0	204.0
3	Grishin *et al.* (1991)	2594	363–417	Central Kazakhstan	58.0	208.0
4	Grishin *et al.* (1991)	2594	377–417	Chingiz-Tarbagay anticlinorium, central Kazakhstan	−5.0	295.0
5	Pechersky & Didenko (1995)	3045	377–417	Arkalyk Mountain, central Kazakhstan	−10.7	87.2
6	Grishin *et al.* (1991)	2594	363–417	Chingiz–Tarbagay anticlinorium, central Kazakhstan	26.0	297.0
7	Pechersky & Didenko (1995)	3045	391–417	Zhalair Range, Yalta Village, central Kazakhstan	−6.8	316.0
8	Pechersky & Didenko (1995)	3045	419–428	Zhanbazar, Ulken-Dogolan Mts, central Kazakhstan	26.2	297.2
9	Pechersky & Didenko (1995)	3045	423–458	Tkenekty and Ushkyzyl Mts, central Kazakhstan	−20.5	52.8
10	Grishin *et al.* (1991)	2594	428–458	Central Kazakhstan	−21.0	63.0
11	Turmanidze (1991)	2852	428–470	Pirnozar, Ushkizi and Tkenekty Mts, mid-Kazakhstan	−16.0	44.8
12	Grishin *et al.* (1991)	2594	443–458	Maykain–Kyziltas zone, central Kazakhstan	−8.0	146.0
13	Grishin *et al.* (1991)	2594	443–458	Chingiz–Tarbagay anticlinorium, central Kazakhstan	38.0	264.0
14	Turmanidze *et al.* (1991)	2626	449–458	mid-Kazakhstan	−21.0	95.0
15	Pechersky & Didenko (1995)	3045	443–470	Chingiz–Tarbagatay anticlinorium, central Kazakhstan	−39.0	78.3
16	Grishin *et al.* (1991)	2594	458–470	Central Kazakhstan	−22.0	51.0
17	Grishin *et al.* (1991)	2594	458–470	Chingiz–Tarbagay anticlinorium, central Kazakhstan	39.0	251.0
18	Turmanidze *et al.* (1991)	2626	458–470	Mid-Kazakhstan	−30.0	60.0
19	Turmanidze *et al.* 1991	2626	458–470	Mid-Kazakhstan	−18.0	103.0
20	Grishin *et al.* (1991)	2594	458–470	Maykain–Kyziltas zone, central Kazakhstan	−14.0	150.0
21	Pechersky & Didenko (1995)	3045	458–470	Itmurundy, Zhangyztan and Ushtogan Mts, Kazakhstan	−24.7	50.1
22	Pechersky & Didenko (1995)	3045	443–495	Karaulcheku allochthon, central Kazakhstan	−21.4	95.5
23	Grishin *et al.* (1991)	2594	458–495	Central Kazakhstan	9.0	347.0
24	Grishin *et al.* (1991)	2594	458–495	Central Kazakhstan	−22.0	61.0
25	Pechersky & Didenko (1995)	3045	458–495	Agyrek Mt, central Kazakhstan	−11.1	149.0
26	Pechersky & Didenko (1995)	3045	458–495	Tolpak Mt, central Kazakhstan	−18.1	103.0
27	Bazhenov *et al.* (2003)		Bashkirian	North Tian-Shan	53.8	202.1
28	Bazhenov *et al.* (2003)		Visean–Serpukhovian	North Tian-Shan	61.6	287.2
29	Bazhenov *et al.* (2003)		Ashgillian	North Tian-Shan	37.4	281.1
30	Bazhenov *et al.* (2003)		Late Caradocian	North Tian-Shan	38.2	246.5
31	Bazhenov *et al.* (2003)		Early Caradocian	North Tian-Shan	31.1	214.7
32	Levashova *et al.* (2003)		Middle Devonian	Chingiz Range	−11.3	85.1
33	Levashova *et al.* (2003)		Early Silurian	Chingiz Range	−33.6	32.7
34	Collins *et al.* (2003)		Early Ordovician	Chingiz Range	−43.9	127.8
35	Collins *et al.* (2003)		Late Cambrian	Chingiz Range	−27.0	168.2
36	This study see Table 2)		Post mid-Devonian	South Kazakhstan	65.6	265.8
37	This study (see Table 2)		Silurian–Early Devonian	South Kazakhstan	56.7	279.3
38	This study (see Table 2)		Post-Late Ordovician	South Kazakhstan	59.8	174.6
39	This study (see Table 2)		Post-Late Ordovician	South Kazakhstan	38.3	145.3
40	This study (see Table 2)		Early–Late Ordovician	South Kazakhstan	37.7	243.3

PLat, PLong: latitude and longitude of the palaeopole.

26.9, $\alpha_{95} = 15.0^\circ$) and (b) for the Silurian and Lower Devonian rocks in the south Chu-Yili Mountains ($D = 346.9^\circ$, $I = 23.8^\circ$, $k = 20.8$, $\alpha_{95} = 12.4^\circ$). This defines a palaeolatitude for the Lower Palaeozoic Stepnyak–north Tian-Shan microcontinent at $8.6^\circ - 8.7^\circ / + 7.6^\circ$ (presumably southerly) for Ordovician time, and a palaeolatitude of the Kazakhstan composite microcontinent at $12.4^\circ + 7.7^\circ / - 6.6^\circ$ (presumably northerly) for Early and mid-Devonian time. Northern latitude, however, cannot be excluded for the Ordovician and resolving this uncertainty needs more detailed study.

(2) Post-folding magnetizations were isolated (a) for the Lower Ordovician rocks ($D = 249.7^\circ$, $I = -57.3^\circ$, $k = 230.4$, $\alpha_{95} = 5.1^\circ$), (b) for the mid-Ordovician rocks ($D = 215.7^\circ$, $I = -51.3^\circ$, $k = 55.4$, $\alpha_{95} = 5.9^\circ$) and (c) for the Silurian and Lower Devonian rocks ($D = 355.5^\circ$, $I = 35.6^\circ$, $k = 18.1$, $\alpha_{95} = 9.6^\circ$). In cases (a) and (b)

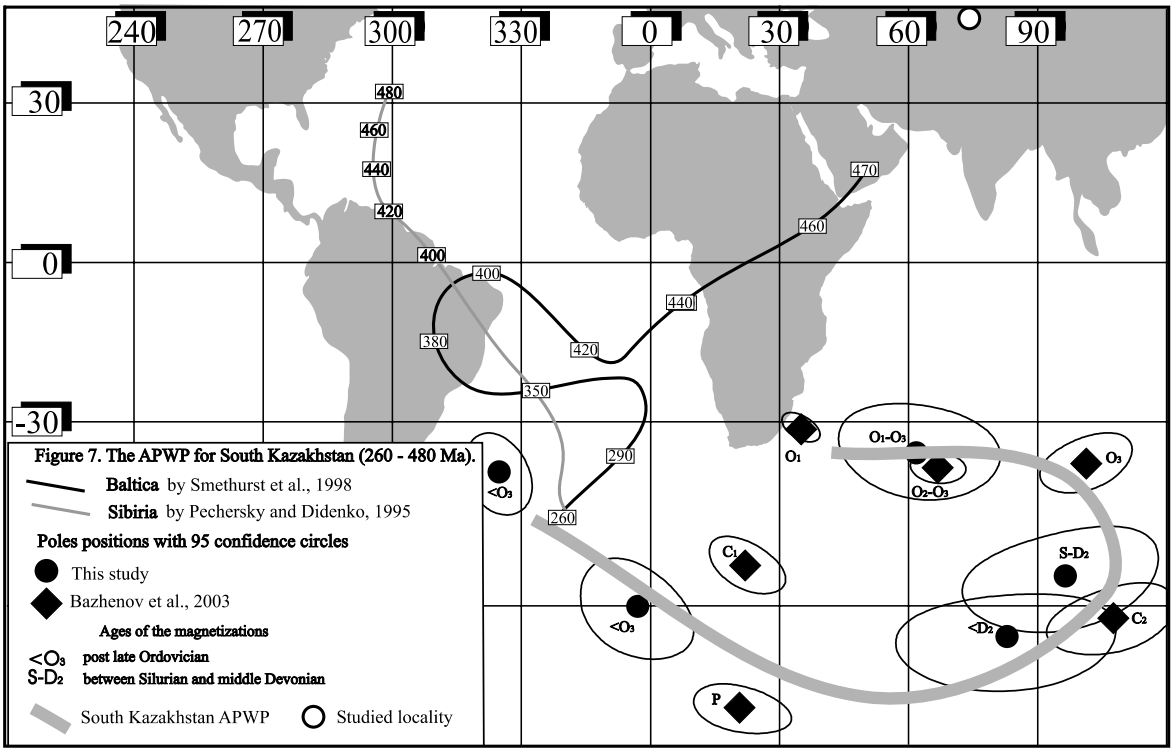


Figure 7. The APWP for south Kazakhstan (260–480 Ma).

the age of magnetization is post-Late Ordovician. In case (c) it is post-Middle Devonian in age.

(3) During the Ordovician to Late Palaeozoic Kazakhstan had good latitude agreement with Baltica and experienced counter-clockwise rotation with respect to both Baltica and Siberia.

(4) In order to decide whether oroclinal bending played a major role in the formation of the Palaeozoic structures in Kazakhstan additional high-quality palaeomagnetic data are needed, especially for northern and central Kazakhstan.

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